Return your solutions on Moodle before **Sunday 29.3. at 23:59**. Your solutions must include intermediate steps and explain your reasoning.

Exercise 4.1. Approximate the integral

$$\int_1^7 \frac{1}{x}\,\mathrm{d}x$$

by hand with (a) the midpoint rule and (b) the trapezoidal rule based on three subintervals ($n = 3$). Draw a picture that illustrates the approximations and compute relative errors of the approximations.

Solution. The true integral is

$$\int_1^7 \frac{1}{x}\,\mathrm{d}x = \cdots = \ln(7) \approx 1.95$$

Observe that

$$a = 1, \quad b = 7, \quad \frac{b-a}{n} = \frac{7-1}{3} = 2,$$

therefore

$$x_i = a + i \cdot \frac{b-a}{n} = 1 + 2i \qquad \text{for} \qquad i \in \{0, 1, \ldots, n\}.$$

(a) Recall the midpoint rule

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{n} \sum_{i=1}^{n} f\left(\frac{x_i + x_{i-1}}{2}\right).$$

The midpoints are

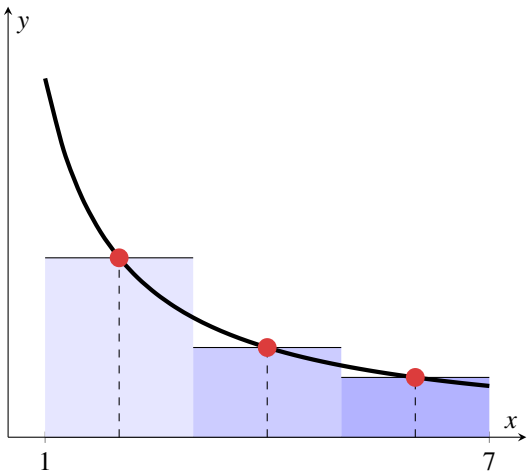
$$\frac{x_i + x_{i-1}}{2} = \frac{1 + 2i + 1 + 2(i-1)}{2} = 2i$$

Therefore

$$\begin{aligned}\int_1^7 \frac{1}{x}\,\mathrm{d}x &\approx 2\sum_{i=1}^{n} f(2i) = 2\Big(f(2) + f(4) + f(6)\Big) \\ &= 2\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6}\right) = 2\frac{6+3+2}{12} \\ &= \frac{11}{6} = 1.8\bar{3}\end{aligned}$$

The relative error is therefore

$$\left|\frac{\ln(7) - 1.8\bar{3}}{\ln(7)}\right| \approx 5.79\%$$



(b) Recall the trapezoidal rule

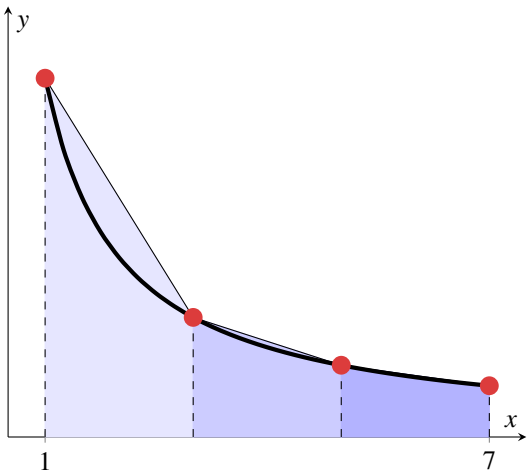
$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{n}\sum_{i=1}^{n}\frac{f(x_i)+f(x_{i-1})}{2}$$

Therefore

$$\begin{aligned}\int_1^7 \frac{1}{x}\,\mathrm{d}x &\approx 2\sum_{i=1}^{n}\frac{f(1+2i)+f(1+2(i-1))}{2} = \frac{2}{2}\Big(f(1)+2f(3)+2f(5)+f(7)\Big)\\ &= \frac{1}{1}+\frac{2}{3}+\frac{2}{5}+\frac{1}{7} = \frac{105+70+42+15}{105}\\ &= \frac{232}{105} \approx 2.21\end{aligned}$$

The relative error is therefore

$$\left|\frac{\ln(7)-232/105}{\ln(7)}\right| \approx 13.55\%$$



□

Exercise 4.2. Suppose that we are to approximate the integral

$$\int_0^1 \exp(\sin(x))\,\mathrm{d}x$$

with the trapezoidal rule. Use Theorem 4.10 to find n such that the trapezoidal rule approximates the integral to 5 decimal places.

Solution. Let

$$E(n) = \left| \int_a^b f(x)\,\mathrm{d}x - \frac{b-a}{n} \sum_{i=1}^{n} \frac{f(x_i)+f(x_{i-1})}{2} \right|$$

denote the absolute error of the trapezoidal rule. Recall that an approximation $\tilde{x}$ to x is said to *approximate x to d decimal places* if

$$|x-\tilde{x}| < \frac{1}{2}10^{-d}.$$

We are interested to find n such that

$$E(n) < \frac{1}{2}10^{-5},$$

where $f(x) = \exp(\sin(x))$, with $a = 0$ and $b = 1$.

Let's recall Theorem 4.10. Suppose $f\colon [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and twice differentiable on (a,b). If there is a constant $L \geq 0$ such that $|f''(x)| \leq L$ for all $x \in (a,b)$, then the trapezoidal rule satisfies

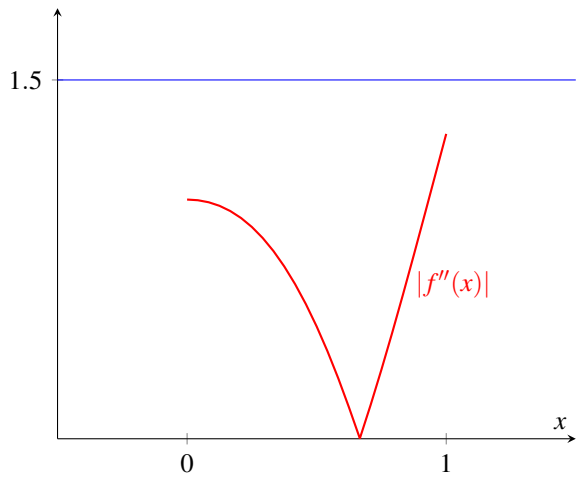
$$E(n) \leq \frac{(b-a)^3 L}{12n^2},$$

where $(b-a)/n = x_i - x_{i-1}$ is the length of each subinterval $[x_{i-1}, x_i]$.

Therefore we can use the fact that

$$\text{if} \qquad \frac{(b-a)^3 L}{12n^2} < \frac{1}{2}10^{-5} \qquad \text{then} \qquad E(n) < \frac{1}{2}10^{-5}.$$

Note that $f''(x) = \frac{d^2}{dx^2}\exp(\sin(x)) = \cdots = \exp(\sin(x))(\cos^2(x) - \sin(x))$. Therefore $|f''(x)|$ is a continuous function on the interval $[a,b]$, by the *Bolzano–Weierstrass theorem* we know that $|f''(x)|$ admits a maximum on $[a,b]$ and therefore there exist infinitely many constants L such that $|f''(x)|$ is bounded from above. The easiest approach to find a suitable L is by plotting the graph of $|f''(x)|$ over the interval $[0,1]$, that is



we observe that $|f''(x)| < 1.5$ for all $x \in [0,1]$, therefore we can set $L = 1.5$ (we could have also chosen $L = 1.3$, $L = 1.7$, $L = 2, \ldots$ or any other $L \geq |f''(1)| \approx 1.2748$, i.e. greater or equal than the maximum) so that $|f''(x)| < L$ is bounded from above for all $x \in (0,1)$. Now the assumptions for Theorem 4.10 are met. Therefore

$$E(n) \leq \frac{(b-a)^3 L}{12n^2} = \frac{(1-0)^3 1.5}{12n^2} = \frac{1}{8}n^{-2}.$$

We can now solve for n such that

$$\begin{aligned}
\frac{1}{8}n^{-2} &< \frac{1}{2}10^{-5} \\
n^{-2} &< 4 \cdot 10^{-5} \\
n^2 &> \frac{1}{4}10^5 \\
n &> \frac{1}{2}10^{\frac{5}{2}} \approx 158.114 \\
n &\geq 159
\end{aligned}$$

Therefore we can conclude that $E(159) < \frac{1}{2}10^{-5}$ for $f(x) = \exp(\sin(x))$, with $a = 0$ and $b = 1$. Note that, since we didn't use the tightest upper bound L for $|f''(x)|$, $n = 159$ is not necessarily the smallest possible solution n s.t. $E(n) < \frac{1}{2}10^{-5}$.

□

Exercise 4.3. Prove Theorem 4.9. That is, prove that the trapezoidal rule is exact for all polynomials of degree at most one. You are not allowed to use Theorem 4.10.

Solution. Every polynomial of degree at most one takes the form $f(x) = \alpha + \beta x$ for some constants $\alpha, \beta \in \mathbb{R}$. We want to show that

$$\int_a^b f(x)\,\mathrm{d}x = \frac{b-a}{n}\sum_{i=1}^{n}\frac{f(x_i)+f(x_{i-1})}{2}$$

for such f. The integral is

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b (\alpha+\beta x)\,\mathrm{d}x = (b-a)\alpha + \frac{(b^2-a^2)}{2}\beta. \tag{1}$$

Because $x_i = a + i(b-a)/n$, the trapezoidal rule is

$$\begin{aligned}
\frac{b-a}{n}\sum_{i=1}^{n}\frac{f(x_i)+f(x_{i-1})}{2} &= \frac{b-a}{2n}\sum_{i=1}^{n}\left[\alpha+\beta\left(a+i\tfrac{b-a}{n}\right)+\alpha+\beta\left(a+(i-1)\tfrac{b-a}{n}\right)\right] \\
&= \frac{b-a}{2n}\sum_{i=1}^{n}\left[2\alpha+\beta\left(a+i\frac{b-a}{n}+a+(i-1)\frac{b-a}{n}\right)\right] \\
&= (b-a)\alpha + \frac{b-a}{2n}\beta\sum_{i=1}^{n}\left[2a+2i\frac{b-a}{n}-\frac{b-a}{n}\right] \\
&= (b-a)\alpha + \frac{b-a}{2n}\beta\left(2na-(b-a)+2\frac{b-a}{n}\sum_{i=1}^{n}i\right)
\end{aligned}$$

Recall then the arithmetic sum formula $\sum_{i=1}^{n} i = n(n+1)/2$, which yields

$$\begin{aligned}
\frac{b-a}{n}\sum_{i=1}^{n}\frac{f(x_i)+f(x_{i-1})}{2} &= (b-a)\alpha + \frac{b-a}{2n}\beta\left(2na-(b-a)+2\frac{b-a}{n}\cdot\frac{n(n+1)}{2}\right)\\
&= (b-a)\alpha + \frac{b-a}{2n}\beta\Big(2na-(b-a)+(b-a)(n+1)\Big)\\
&= (b-a)\alpha + \frac{b-a}{2n}\beta\Big(2na+n(b-a)\Big)\\
&= (b-a)\alpha + \frac{b-a}{2}\beta(2a+b-a)\\
&= (b-a)\alpha + \frac{b-a}{2}\beta(b+a)\\
&= (b-a)\alpha + \frac{b^2-a^2}{2}\beta.
\end{aligned}$$

This equals (1). Thus the trapezoidal rule is exact for all functions of the form $f(x) = \alpha + \beta x$. □

Exercise 4.4. Implement the midpoint rule and the trapezoidal rule in a programming language of your choice. Use them to approximate the integrals

$$\int_{-1}^{1}\exp(-x^2)\,\mathrm{d}x = \sqrt{\pi}\cdot\operatorname{erf}(1) \approx 1.4936482656248540507989 34\ldots,$$

where erf is called the *error function*, and

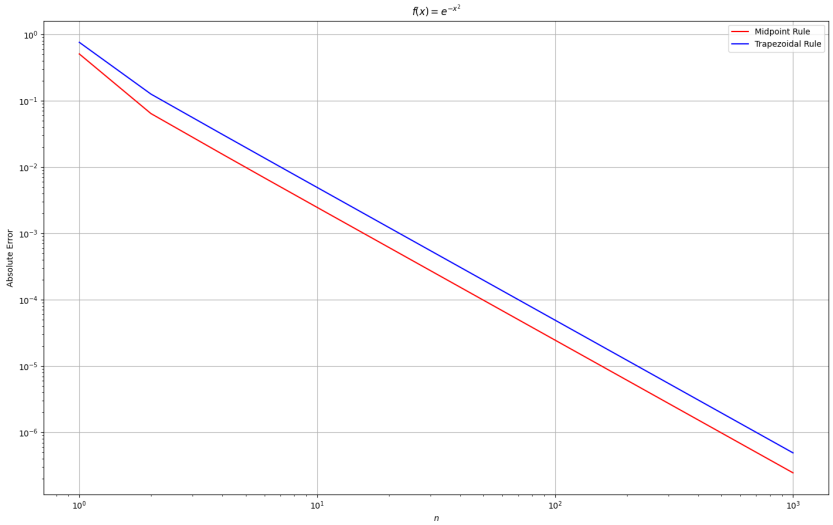
$$\int_{-1}^{1}\exp(-|x|)\,\mathrm{d}x = 2-\frac{2}{e}$$

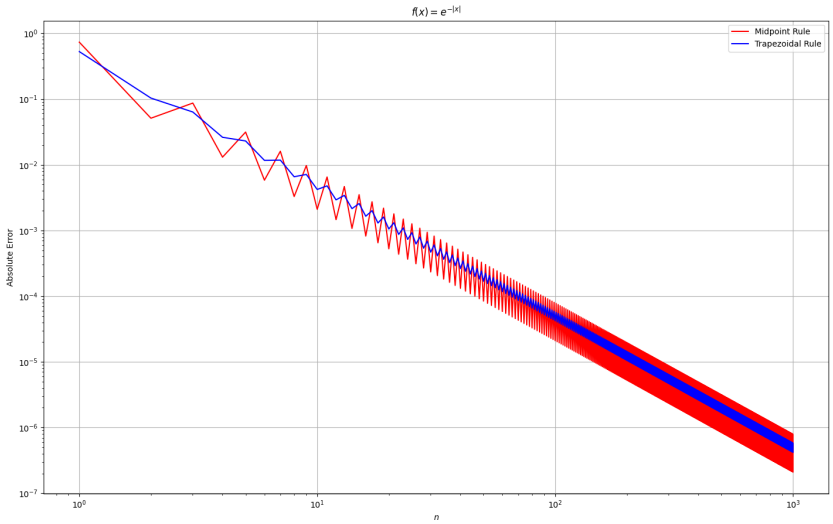
with $n \in \{1,2,\ldots,1000\}$. Plot the absolute errors on a log-log plot.

Solution.

```
from math import e, erf, pi, sqrt
from functools import partial

import numpy as np
import matplotlib.pyplot as plt


def quadrature(f, t, w):
    return f(t) @ w


def midpoint(f, n, a, b):
    x = np.linspace(a, b, n + 1)
    t = (x[1:] + x[:-1]) / 2 # midpoints
    h = (b - a) / n
    return quadrature(f, t, np.full(n, h))


def trapezoidal(f, n, a, b):
```

```
    x = np.linspace(a, b, n + 1)
    h = (b - a) / n
    w = np.r_[h / 2, np.full(n - 1, h), h / 2]
    return quadrature(f, x, w)


# dynamic plot settings
midpoint.color = 'r'
trapezoidal.color = 'b'
for quadfunc in (midpoint, trapezoidal):
    quadfunc.label = (quadfunc.__name__ + " rule").title()


# constants
N = np.r_[1 : 1000 + 1]
a = -1
b = +1


exercises = list()

f = lambda x: np.exp(-(x**2))
I = sqrt(pi) * erf(1) # excat integral

f.name = r"$f(x) = e^{-x^2}$"
exercises.append((f, I))

f = lambda x: np.exp(-np.abs(x))
I = 2 - 2 / e # excat integral

f.name = r"$f(x) = e^{-x}$"
exercises.append((f, I))

# solve

for f, I in exercises:
    plt.figure()

    for quadfunc in (midpoint, trapezoidal):
        _quadfunc = partial(quadfunc, f, a=a, b=b)

        approximations = np.fromiter(map(_quadfunc, N), dtype=float)
        err_n = np.abs(I - approximations)

        plt.plot(N, err_n, quadfunc.color, label=quadfunc.label)

    plt.grid(True)
    plt.legend()
    plt.title(f.name)
    plt.ylabel('Absolute Error')
    plt.xlabel(r'$n$')
    plt.yscale('log')
    plt.xscale('log')
    plt.show()
```





□

Exercise 4.5. The indefinite integral of the density function,

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2},$$

of the standard normal distribution does not admit a closed-form expression in terms of a finite number of elementary functions. Consequently, the values of the cumulative distribution function

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{1}{2}t^2} \, \mathrm{d}t$$

are tabulated for different values of x in various sources. Implement Simpson's rule in a programming language of your choice and use it to complete the following table:

x	$\Phi(x)$
0.0	?
0.2	?
0.4	?
0.6	?
$\vdots$	$\vdots$
3.0	?
3.2	?
3.4	?

The first column contains values of x. You can freely select the points for Simpson's rule *as long as your approximations for the cumulative distribution function increase with x*. That is, your approximations should satisfy $\Phi(x) > \Phi(z)$ when $x > z$. You are allowed to use the fact that

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} e^{-\frac{1}{2}t^2} \, \mathrm{d}t = \frac{1}{2}.$$

Solution.

```
from itertools import accumulate

import numpy as np


def quadrature(f, t, w):
    return f(t) @ w


f = lambda x: np.exp(-1 / 2 * x**2) / np.sqrt(2 * np.pi)

step = 0.2
x = np.r_[0 : 3.4 + step : step]

a = x[:-1]
b = x[1:]

# Simpson's rule
t = np.c_[a, (a + b) / 2, b]
w = np.r_[1, 4, 1] * step / 6

areas = quadrature(f, t, w)
I = np.fromiter(accumulate(areas, initial=1 / 2), dtype=float)

print(np.c_[x, I])
```

x	$\Phi(x)$
0.0	0.50000000
0.2	0.57925984
0.4	0.65542198
0.6	0.72574718
0.8	0.78814491
1.0	0.84134502
1.2	0.88493053
1.4	0.91924346
1.6	0.94520075
1.8	0.96406966
2.0	0.97724981
2.2	0.98609647
2.4	0.99180238
2.6	0.99533874
2.8	0.99744481
3.0	0.99865006
3.2	0.99931283
3.4	0.99966305

□